Karthick Raja

Data Scientist,
Philips Research

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Social Network

karthickrajas

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Framework

Tensorflow, Pytorch, Python, R, Statistis, Operations, GIT, RDBMS, Vertica, NLP, ComputerVision

Hard Skills

- Programming for Data Science
- Experience with databases
- Data visualisation for business
- Big data framework
- Understanding of optimization

Soft Skills

- Business Acumen ● ● ● ●
  - Worked with multiple non Analytics teams
- Effective communication ● ● ● ●
  - Youth Leadership Club,Philips

Work Experience

July'19 - Now — **Data Scientist** — Philips Research
Working on precision diagnosis and predictive maintenance projects

Jan-Jun '19 — **Data Scientist Intern** — CITI India Services Limited
Worked on contact strategy optimization and ad placement for web applications using Reinforcement Learning

Apr-Jun '18 — **Data Scientist Intern** — GE Digital
Natural language processing and Computer vision projects

Jun'15-July'17 — **System Engineer** — Tata Consultancy Services
Worked with media research client for viewership analytics and report generation

Education

2019 — **Masters in Business Analytics** — Indian Institute of Science
CGPA: 8.1 | Classical and Advanced Machine learning, Data Mining, Computer vision, Natural language processing and Reinforcement Learning

Master Thesis — Citi Services India Limited
Ad Personalisation using Contextual Multi Arm Bandits: Dynamic algorithm to target customer with Ads specific to their interest

2017 — **Bachelor of Engineering** — TCE, Anna university
CGPA: 9.2 | Civil engineering, specialised on Analysis and Design of steel structures, Numerical modelling and programming

Bachelor Thesis — funded by Steel Research Institute: INSDAG
Experimental study of Hot Rolled Square Hollow Sections: Modelling failure of steel columns to study the load carrying capacity of columns and study their actual performance

Publications

2020 — **AI in Banking and Finance Industry**
Smart Technologies: Scope and Applications (Springer-review)

2019 — **Abstractive Text Summarization**
Smart Technologies: Scope and Applications (Springer-review)

2018 — **Data Modelling and Micro Targeting of voters for Elections**
National Conference on strategic decision making using BA

2015 — **Experimental study on Hot rolled square hollow section**
International Journal on Mechanical, Civil and Control Engineering

Achievements and Awards

2019	Attended winter school at clausthal University of Tech	Masters, IISc
2019	Won Citi Ideas Challenges	Citi Services
2018	Won GE DnA Hackathon	GE Digital
2017	First prize in National Conference on strategic decision making and business analytics	Masters, IISc
2015	ILP Kudos, Award for meritorious trainee of the batch	TCS

Extra-Curricular Activities

Blogging	Writes personal as well as internal (philips) technical blogs (karthickrajas.github.io)
Swimming	Recently learn skill and a exercise routine
Clubs	Young Leaders Club - Philips Analytics Society of India, Bangalore

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About Me

I believe, along with the technical knowledge and coding prowess, one should possess general business understanding and industry specific knowledge to become a successful data scientist. This can help in impact-ful decision making and enrich customer experience.

Interests

- Computer vision
- NLP
- Federated learning
- Reinforcement learning

Languages

- English ●●●●●
- Hindi ●●●○○
- Tamil ●●●●●

Memberships

- IISc Alumni Association
- Analytics Society of India

Industry Projects

Deep Similar Image Retrieval

Aim	Develop a PoC for Similar Image Retrieval, for Bio-Medical images
outcome	Cloud perform feature for similar image retrieval
Methods	Experimented with several Classical **CBIR** feature extraction techniques, **Auto encoder** and **Siamese neural network**

Liver and Lesion Segmentation, LIRADS classification

Aim	Develop deep learning model for segmentation of liver and cancerous nodules and score them for Heptocellular Carcinoma
outcome	Model with high DICE score, Currently working on the LIRADS score
Methods	Experimented with **U-NET** and **U-NET with attention mechanism** for pixel level segmentation of liver and lesion with several losses

Predictive Maintenance

Aim	Develop models to predict the failure of several parts of Diagnostic resources including Ultrasound, X-Ray machine
outcome	Multiple models of different nature predicting the current state (defective) of the machine and time to failure of parts
Methods	Have analysed and implemented several **threshold based models** and classical machine learning models like **decision trees** and **SVM**

Model-Concept Drift

Aim	Develop mechanism to identify concept drift in developed models and Develop Adaptive Models to overcome concept drift
outcome	Have developed a test bed to simulate model drift and working on adaptive hoeffding trees models to improve model performance
Methods	**ADWIN** -Adaptive windows , **Adaptive Hoeffding Trees**, **Concept adaptive very fast decision trees (CVFDT)**

Classification of Power plants using Satelite pictures

Aim	Develop computer vision model for classification of power plant imagery from satellite surveillance
outcome	Feature in a cloud based tool for identifying powerplants
Methods	Employed **Deep CNN** for the unbalanced classification, We have utilized **Transfer learning** since we have very less image for classification

Text Summarization

Aim	To develop an effective summariser for contractor documents submitted with machinary maintenance
outcome	Summarizer tool to be part of the cloud environment assessing documents of contractors. Developed POC for abstractive summarization on summarizing the contents available in the documents
Methods	For extractive purposes, i experimented with **TextRank** and for abstractive:**Seq2Seq** Deep learning architecture for used on **LSTM** and **GRU** memory cells

Research Projects

Personalizing ads for online customers - Reinforcement Learning

Aim	To come up with an online learning model for showing ads to customer and improve cross selling of banking products
outcome	Reinforcement learning model for recommending products based on user interactions
Methods	**Thompson sampling** for context-free recommendation and **LinUCB - Linear Upper Confidence Bound** for contextual recommendation and **Policy Gradients** for complete Reinforcement Learning solution